Engine Model(s): PT6A

72-10-00

REDUCTION GEARBOX - MAINTENANCE PRACTICES

1. General

A. Maintenance personnel should make reference to the INTRODUCTION section and Chapter 70-00-00 STANDARD PRACTICES of this manual to familiarize themselves with general procedures.

B. Install suitable protective caps/covers over all disconnected tubes/lines and component openings.

C. Lockwire used, shall unless otherwise specified, comply with specification AMS 5687, heat and corrosion resistant steel wire MS9226-03, which is 0.025 inch diameter, and will not be specified in instructions.

2. Consumable Materials

The consumable materials listed below are used in the following procedures.

Item No.	Name
PWC03-001	Oil, Engine Lubricating
PWC05-037	Enamel, Aluminum Epoxy
PWC05-061	Cloth, Abrasive
PWC05-101	Cloth, Abrasive
PWC05-270	Paper, Abrasive
PWC07-023	Coating, Aluminum and Zinc
PWC07-062A	Coating-Base, Aluminum
PWC07-062B	Coating-Base, Aluminum
PWC07-062C	Coating-Base, Aluminum
PWC08-002	Adhesive, Fluorosilicone
PWC09-001	Sealant, Silicone Rubber
PWC11-014	Alcohol, Isopropyl
PWC11-016	(Use PWC11-014)
PWC11-027	Solvent, Petroleum
PWC11-031	Cleaner, Engine
PWC13-001	Primer, Epoxy

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Item No.	Name
PWC13-003	Primer, Coating

3. Special Tools

The special tools listed below are used in the following procedures.

Tool No.	Name	Application
PWC57233	Puller	Replaces PWC32072

4. Fixtures, Equipment and Supplier Tools

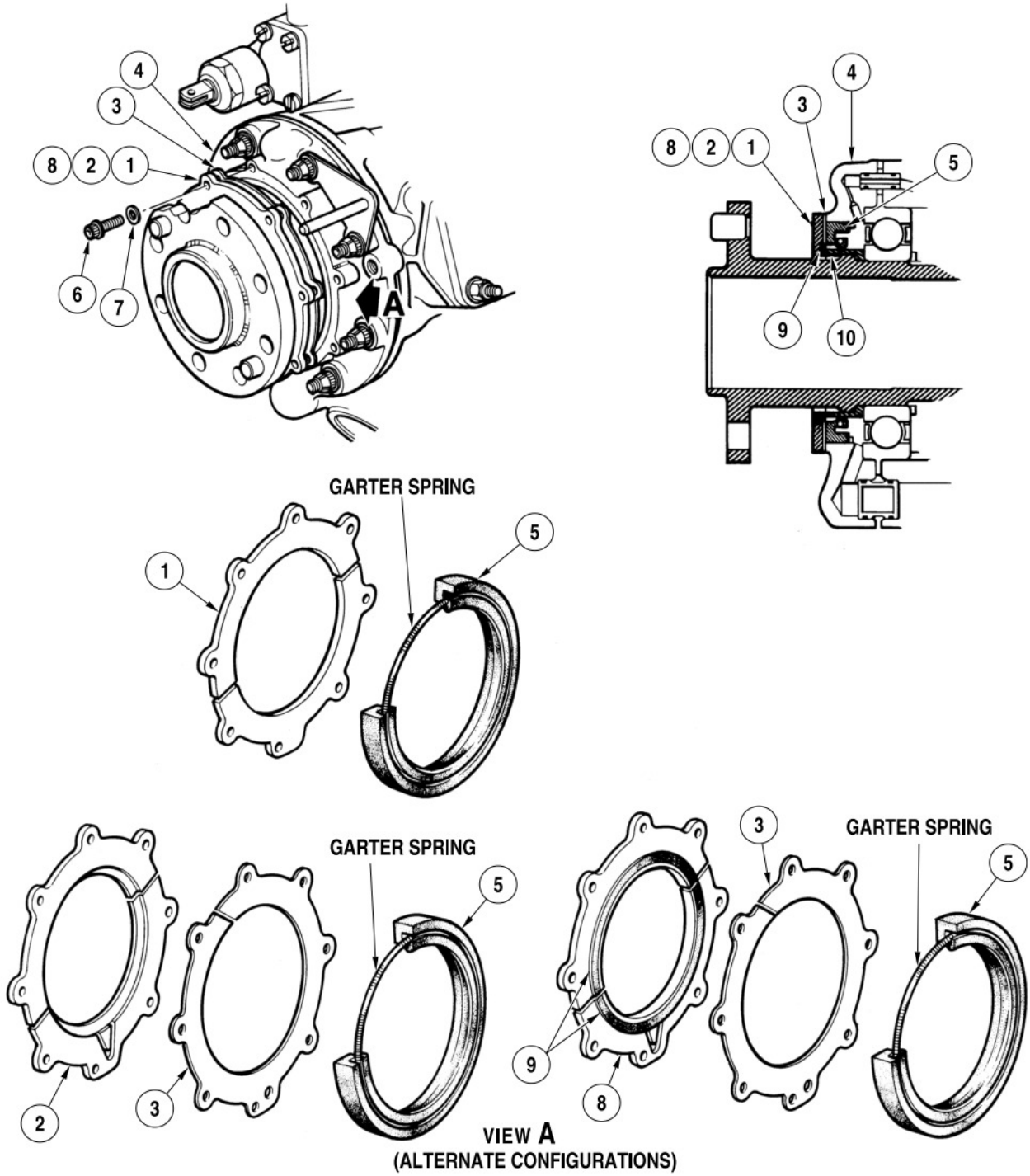
Not Applicable

5. Removal/Installation

A. Removal of Propeller Shaft Oil Seal - Pre-SB1227, Post-SB1377 and Post-SB1381 and Post-SB1652 (Ref. Fig. 201)

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Figure 201 Removal/Installation of Propeller Shaft Oil Seal



Pre-SB1227, Post-SB1377, Post-SB1381 and Post-SB1652

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1. Oil Seal Retaining Ring Halves (Pre-SB1227)
2. Oil Seal Retaining Ring Halves (Post-SB1377)
3. Gasket (Post-SB1377)
4. Thrust Bearing Cover
5. Seal and Garter Spring
6. Bolt
7. Washer
8. Oil Seal Retaining Ring Half (Post-SB1381/Post-SB1652)
9. Felt Strip Insert (Post-SB1381/Post-SB1652)
10. Seal Runner

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(1) Remove the propeller (Ref. Aircraft Maintenance Manual).

(2) Remove eight bolts (6, Fig. 201) and washers (7) that attaches the oil seal retaining ring halves (1, Pre-SB1227), (2, Post-SB1377) or (8, Post-SB1381/Post-SB1652) to the propeller thrust bearing cover (4). Remove the ring halves.

(3) If installed, remove the gasket (3, Post-SB1377) and discard.

(4) Work out the seal (5) from the cavity, taking care not to damage the bore of the thrust bearing cover assembly or the surface of the oil seal runner (10).

(5) Remove the garter spring from the inside diameter rear face of the seal (5). Unwind the spring and remove from the propeller shaft. Remove the seal by stretching over the retaining ring and propeller shaft flange.

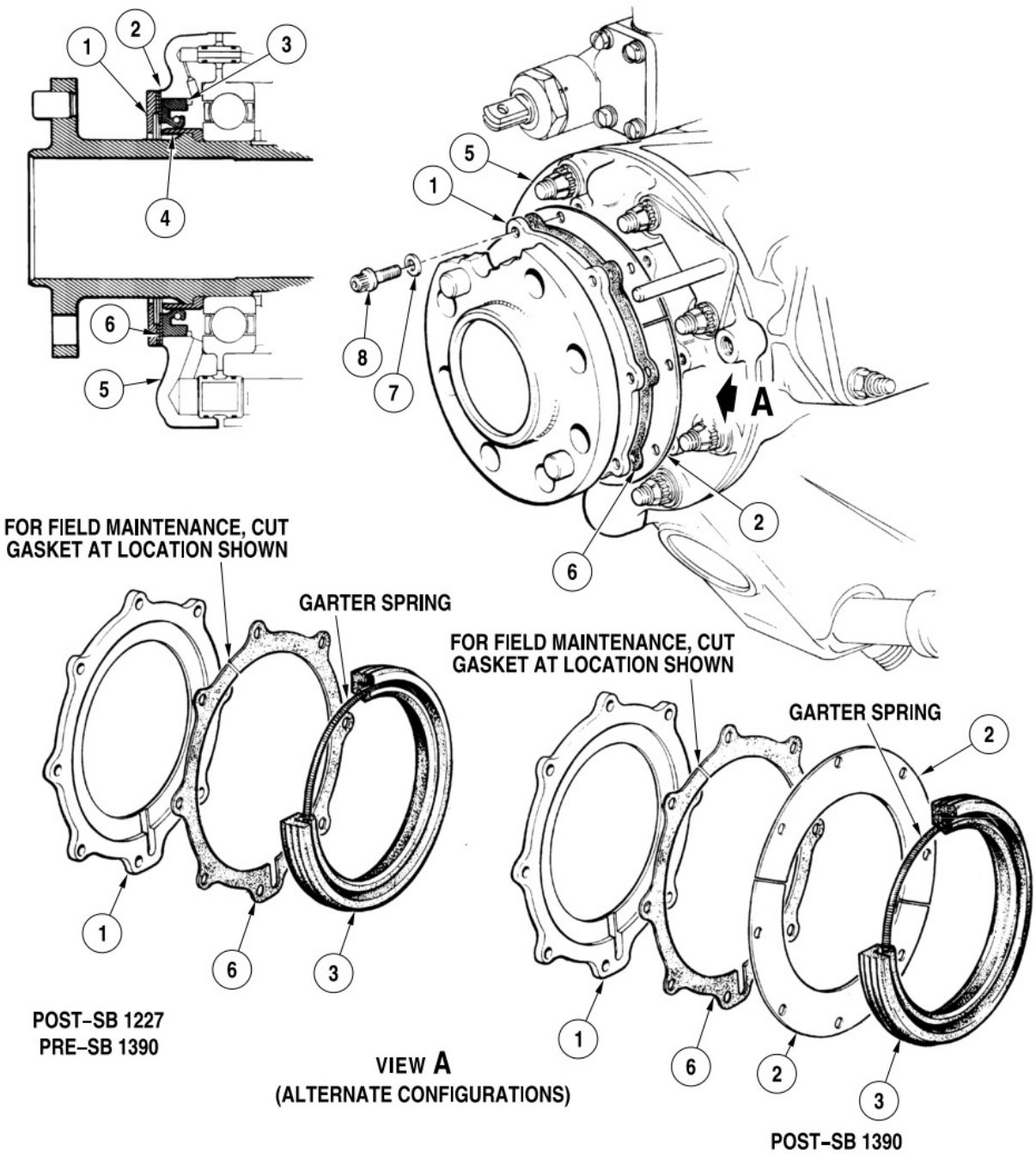
NOTE: The seal may be cut in order to remove it. When cutting, protect the surface of the propeller shaft from possible nicks or scratches caused by the cutting instrument.

(6) Clean the gasket residue from the retaining ring halves and thrust bearing cover mating surfaces using a suitable non-metallic scraper.

B. Removal of Propeller Shaft Oil Seal - Post-SB1227, Pre-SB1390 and Post-SB1390 (Ref. Fig. 202)

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Figure 202 Removal/Installation of Propeller Shaft Oil Seal



Post-SB1227, Pre-SB1390 and Post-SB1390

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1. Oil Seal Retaining Ring
2. Oil Seal Support Ring Halves (Post-SB1390)
3. Seal Element
4. Seal Runner
5. Thrust Bearing Cover
6. Gasket
7. Washer
8. Bolt

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(1) Remove the propeller (Ref. Aircraft Maintenance Manual).

(2) Place a suitable container under the front end of the reduction gearbox and disconnect the airframe drain line from the nipple at the bottom of the thrust bearing cover (5, Fig. 202). Allow any residual oil to drain into the container.

(3) Remove the bolts (8), washers (7) and oil seal retaining ring (1) from the thrust bearing cover. Move the retaining ring (1) forward to be flush against the flange of the propeller shaft. Remove the gasket (6) and discard.

(4) If Post-SB1390 oil seal support ring halves (2) are fitted, note the split line location, identify the top and bottom segments, and remove from the propeller shaft.

(5) Work the seal element (3) from the cavity, taking care not to damage the bore of the thrust bearing cover assembly or the surface of the oil seal runner (4).

(6) Remove the garter spring from the inside diameter rear face of the seal. Unwind the spring and remove from the propeller shaft. Remove the seal by stretching the seal over the retaining ring and propeller shaft flange.

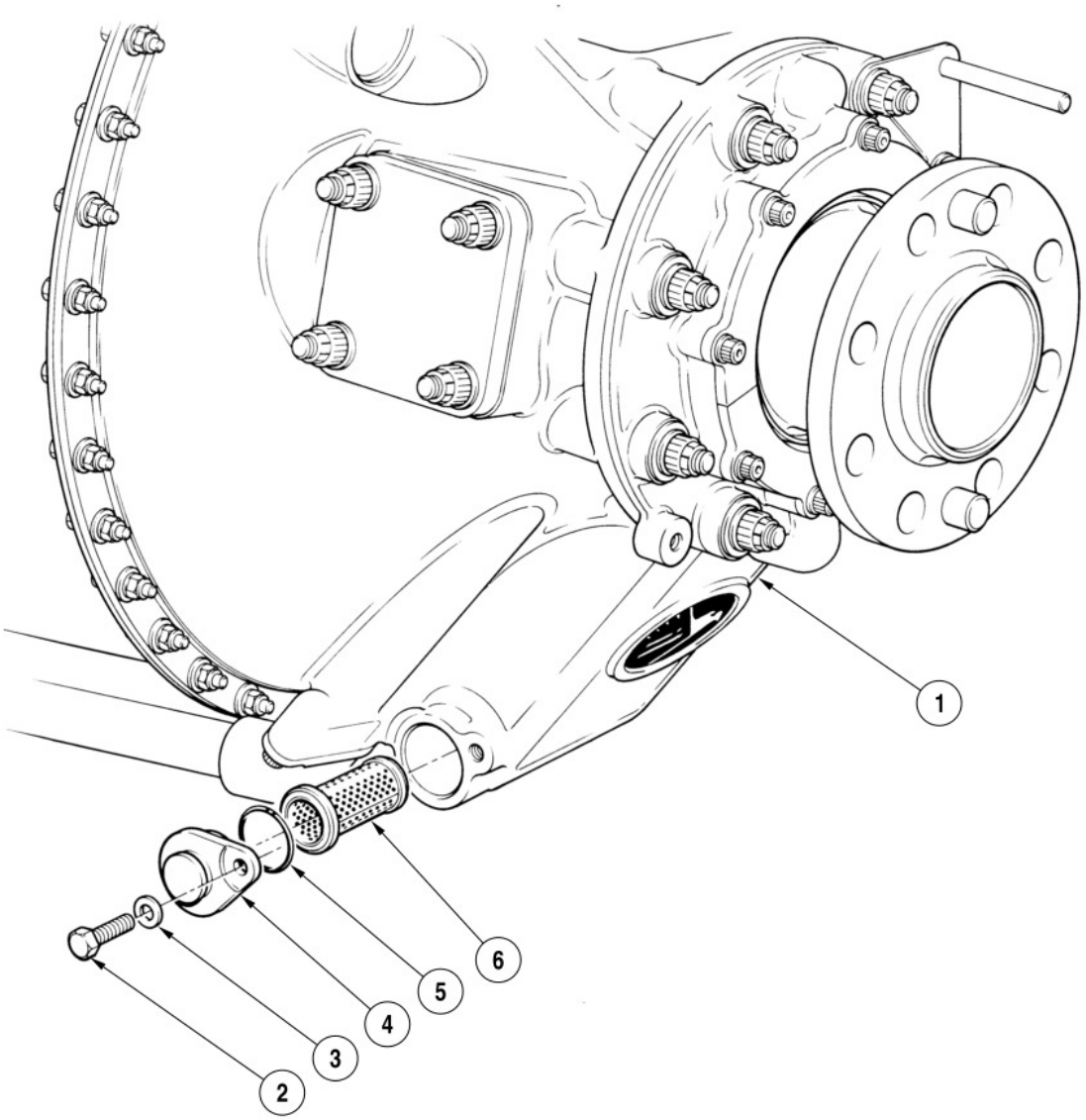
NOTE: The seal may be cut to facilitate removal. When cutting the seal, protect the surface of the propeller shaft from possible nicks or scratches caused by the cutting instrument.

(7) Clean the gasket mating surfaces using a suitable non-metallic scraper.

C. Removal of Reduction Gearbox Oil Strainer (Ref. Fig. 203)

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Figure 203 Removal/Installation of Scavenge Oil Strainer



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1. Reduction Gearbox
2. Bolt
3. Washer
4. Scavenge Oil Strainer Plug
5. Preformed Packing
6. Scavenge Oil Strainer Assembly

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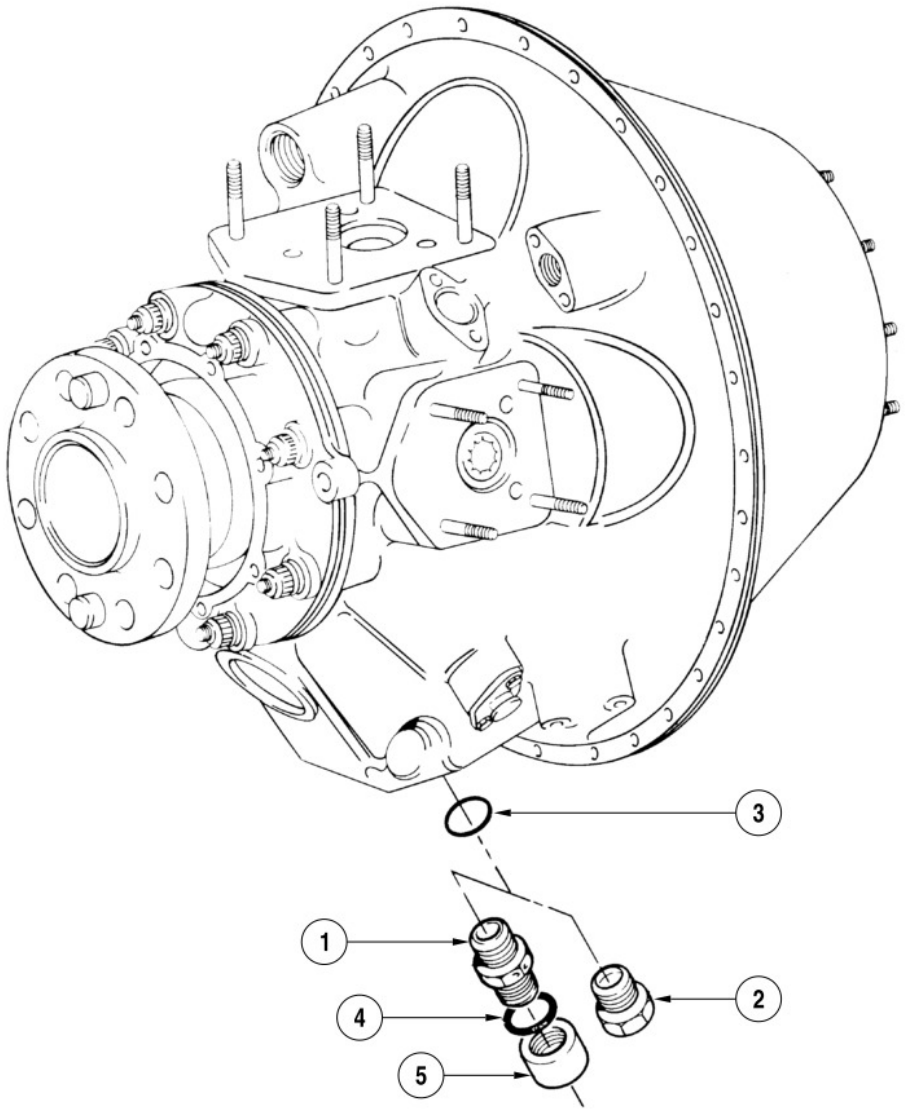
(1) Remove the bolt (2, Fig. 203) and washer (3) that attach the scavenge oil strainer plug (4) to the reduction gearbox (1). Withdraw the strainer plug with puller (PWC57233) and remove the preformed packing (5).

(2) Remove the scavenge oil strainer assembly (6) from the cored passage in the gearbox.

D. Removal of Magnetic Chip Detector - Post-SB1217 (Ref. Fig. 204)

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Figure 204 Magnetic Chip Detector



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1. Chip Detector (Post-SB1217)
2. Plug (Pre-SB1217)
3. Preformed Packing
4. Preformed Packing (Post-SB1217)
5. Cover (Post-SB1217)

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NOTE: On Pre-SB1217 engines, a drain plug was fitted in lieu of a chip detector.

CAUTION: AVOID FORCING OR CONTACT WITH THE OIL PRESSURE TRANSFER ELBOW WHEN YOU REMOVE THE CHIP DETECTOR. DAMAGE TO THE ELBOW MOUNTING LUGS CAN OCCUR RESULTING IN LOSS OF ELBOW RETENTION AND OIL LEAKAGE.

(1) Remove the cover (5, Fig. 204), or disconnect the airframe cable, from the chip detector (1) at the 6 o'clock position on the reduction gearbox. If the cover is removed, discard the preformed packing (4) if the inspection shows deterioration or damage to the packing.

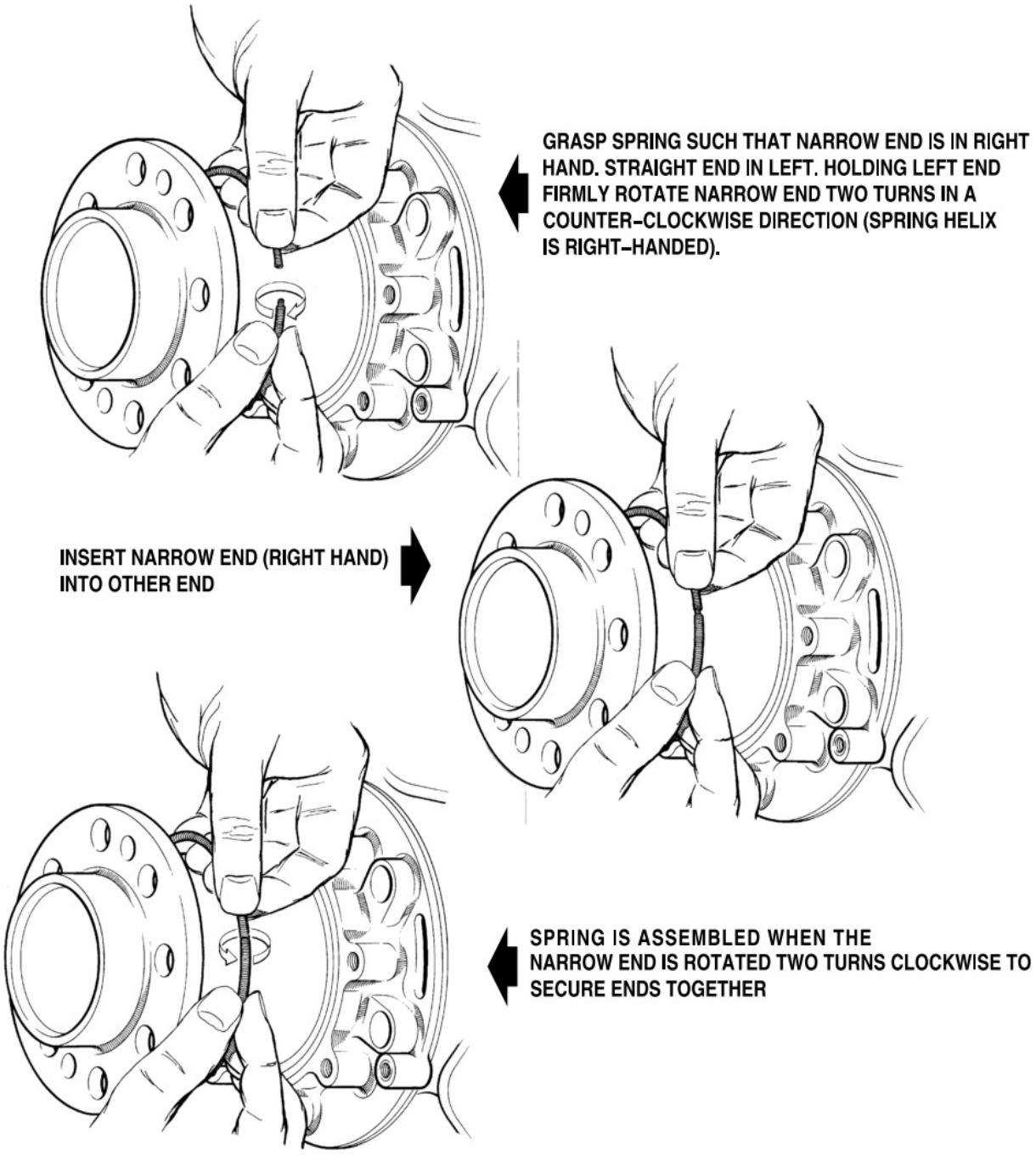
(2) Provide a suitable drip pan under the reduction gearbox to collect any residual oil.

(3) Remove the chip detector (1) from the reduction gearbox and discard the preformed packing (3).

E. Installation of Propeller Shaft Oil Seal - Pre-SB1227, Post-SB1377, Post-SB1381 and Post-SB1652 (Ref. Figs. 201 and 205)

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Figure 205 Installation of Propeller Shaft Oil Seal Garter Spring



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CAUTION: HANDLE THE SEAL WITH CARE TO PREVENT DAMAGE. AVOID APPLYING LOADS LOCALLY TO THE LIP OF THE SEAL, AS THE SEAL IS SOFT AND HAS LOW TEAR RESISTANCE.

(1) Remove the garter spring from the new seal (5, Fig. 201).

(2) Tape a strip of plastic material (polyethylene sheet) over the flange of the propeller shaft and apply a liberal layer of engine oil (PWC03-001). This will let the seal slip over the flange without undue stress.

(3) Preheat the seal to a maximum of 93°C (200°F) for 20 minutes.

(4) Moisten seal (5, Fig. 201) in engine oil (PWC03-001) and install over propeller mounting flange of the propeller shaft with the V-side of the seal facing the thrust bearing cover (4).

(5) Install the garter spring in the V-section of the seal (Ref. Fig. 205).

(6) Gently press the seal and garter spring into the cavity formed by the bore of the thrust bearing cover (4, Fig. 201) and oil seal runner (10). Work the seal into the cavity with thumbs until the seal is correctly seated in the cavity.

(7) Remove the plastic sheet from the propeller shaft and wipe propeller shaft dry of all residual oil. Install the oil seal retaining ring halves (1, Pre-SB1227), (2, Post-SB1377) or (8, Post-SB1381/Post-SB1652) as follows:

CAUTION: MAKE SURE THE REAR FACE OF THE RETAINING RING HALVES ARE CLEAN AND THE SURFACE IS FREE OF ABRASIONS, NICKS, PITTING AND SCORING.

(a) Install Pre-SB1227 retaining ring halves (1) over the oil seal (5) making sure the gap between the two halves is approximately horizontal and the bolt holes are aligned with those on the thrust bearing cover (4).

(b) For Post-SB1381/Post-SB1652 engines, moisten the felt strip inserts (9) on the oil seal retaining ring halves (8) with engine oil (PWC03-001).

CAUTION: THE OIL SEAL SUPPORT GASKET IS VERY BRITTLE; DO NOT OVERSTRESS DURING INSTALLATION.

(c) When installing the Post-SB1377 or Post-SB1381 oil seal assembly, place the pre-split support gasket (3) carefully over the propeller shaft with the split at the one o'clock position and the drain port at bottom centerline. Install the retaining ring halves (2 or 8) so the lower half ring oil drain channel aligns with the cut-out in the gasket and drain spigot. Align the bolt holes of both ring halves with the holes in the thrust bearing cover (4), making sure the joints are at the four and ten o'clock positions (all references looking rearward).

(d) Make sure the Post-SB1381/Post-SB1652 felt strip inserts (9) engage the lip of the seal runner (10).

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CAUTION: BOLTS MUST BE SAFETY WIRED WITH FINISHING PIG TAIL POINTING INBOARD OR TOWARDS PROPELLER SHAFT TO PREVENT DAMAGE TO BETA FEEDBACK RING.

(8) Align and attach the retaining ring halves (1, 2 or 8, as applicable) with washers (7) and bolts (6). Tighten the bolts in diametric sequence, torque 24 to 36 lb.in., and lockwire in pairs.

(9) Install the propeller (Ref. Aircraft Maintenance Manual).

NOTE: The use of Post-SB1377 configuration is an option to Post-SB1390 configuration, not a replacement.

F. Installation of Propeller Shaft Oil Seal - Post-SB1227, Pre-SB1390 and Post-SB1390 (Ref. Figs. 205)

CAUTION: HANDLE THE SEAL WITH CARE TO PREVENT DAMAGE. AVOID APPLYING LOADS LOCALLY TO THE LIP OF THE SEAL, AS THE SEAL IS SOFT AND HAS LOW TEAR RESISTANCE.

(1) Remove the garter spring from the new seal (3, Fig. 202).

(2) Tape a strip of plastic material (polyethylene sheet) over the flange of the propeller shaft and apply a liberal layer of engine oil (PWC03-001). This will let the seal slip over the flange without undue stress.

(3) Preheat the seal to a maximum of 93°C (200°F) for 20 minutes.

(4) Moisten seal (3, Fig. 202) in engine oil (PWC03-001) and install over propeller mounting flange of the propeller shaft with the V-side of the seal facing the thrust bearing cover (5).

(5) Install the garter spring in the V-section of the seal (Ref. Fig. 205).

(6) Gently press the seal and garter spring into the cavity formed by the bore of the thrust bearing cover (5, Fig. 202) and oil seal runner (4). Work the seal into the cavity with thumbs until the seal is correctly seated in the cavity.

(7) Remove the plastic sheet from the propeller shaft and wipe propeller shaft dry of all residual oil.

(8) If the Post-SB1390 oil seal support ring halves (2) are to be installed, position the segments with the split line as noted at disassembly, and with the top and bottom halves properly located against the seal (3).

CAUTION: THE OIL SEAL SUPPORT GASKET IS VERY BRITTLE; DO NOT OVERSTRESS DURING INSTALLATION.

(9) Cut (slit) the new gasket (6) midway between the two holes at a right angle to the vertical centerline, then carefully install on the propeller shaft.

CAUTION: BOLTS MUST BE SAFETY WIRED WITH FINISHING PIG TAIL POINTING INBOARD OR TOWARDS PROPELLER SHAFT TO PREVENT DAMAGE TO BETA FEEDBACK RING.

(10) Align the groove in the retaining ring (1) and corresponding cut-out in the gasket with the

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drain spigot in the thrust bearing cover. Attach the retaining ring with eight bolts (8) and washers (7). Tighten the bolts in diametric sequence, torque 24 to 36 lb.in., and lockwire in pairs.

(11) Install the propeller (Ref. Aircraft Maintenance Manual).

(12) Connect the airframe drain line to the nipple at the bottom of the thrust bearing cover (Ref. Aircraft Maintenance Manual).

G. Installation of Reduction Gearbox Oil Strainer (Ref. Fig. 203)

(1) Locate the strainer (6) in cored passage at the front of the reduction gearbox (1). Make sure that the strainer is pressed fully home.

(2) Moisten the preformed packing (5) with engine oil (PWC03-001) and install in the recess on the boss of the strainer plug (4).

(3) Install the strainer plug on the reduction gearbox and secure with the bolt (2) and washer (3). Tighten the bolt 36 to 40 lb.in., and fasten with lockwire.

H. Installation of Magnetic Chip Detector - Post-SB1217 (Ref. Fig. 204)

NOTE: On Pre-SB1217 engines, a drain plug was fitted in lieu of a chip detector.

(1) Install a new preformed packing (3, Fig. 204) on the chip detector (1).

CAUTION: THE CHIP DETECTOR CAN BE QUITE EASILY OVERTIGHTENED BY MODERATE HAND PRESSURE. USE OF A TORQUE WRENCH IS THEREFORE IMPORTANT TO AVOID OVERTORQUING WHICH COULD RESULT IN FRACTURE OF THE CHIP DETECTOR BODY.

CAUTION: AVOID FORCING OR CONTACT WITH THE OIL PRESSURE TRANSFER ELBOW WHEN YOU INSTALL THE CHIP DETECTOR. DAMAGE TO THE ELBOW MOUNTING LUGS CAN OCCUR RESULTING IN LOSS OF ELBOW RETENTION AND OIL LEAKAGE.

(2) Install the chip detector in the boss at the 6 o'clock position in the reduction gearbox. Tighten 45 to 55 lb.in.

(3) If necessary, install a new preformed packing (4) in the cover (5).

(4) Install the cover on the chip detector, or reconnect the airframe cable, as applicable. If the cover was reinstalled, tighten fingertight only; do not use tools. Lockwire the cover and/or chip detector.

NOTE: On field-modified engines, a circular knurled cover (MS25043-120), with integral gasket and chain, may be fitted in lieu of a drilled hexagon cover. This cover should be reinstalled on the chip detector fingertight only, and the chain suitably fastened with lockwire to the chip detector.

6. Cleaning/Painting

A. Scavenge Oil Strainer Assembly

(1) Soak scavenge oil strainer assembly in clean solvent (PWC11-027) or (PWC11-031). Remove

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oil sludge and examine for evidence of metallic particles. Open up blocked holes using non-metallic scraper.

(2) Wipe clean with lint-free cloth and dry with clean, dry compressed air.

B. Magnetic Chip Detector

CAUTION: DO NOT SOAK DETECTOR IN SOLVENT.

(1) Seal electrical receptacle.

(2) Use a medium stiff-bristle brush or spray to clean with solvent (PWC11-027) or (PWC11-031).

(3) Make sure detector is free of metal particles. Dry with clean, dry compressed air.

(4) If not required for immediate installation, store chip detector in clean polyethylene bag.

C. Reduction Gearbox Rear Housing (Heavy Maintenance Only)

(1) Use petroleum solvent (PWC11-027) or (PWC11-031) to remove any residue from reduction gearbox rear case.

(2) Wipe case clean with a lint-free cloth and dry with clean, dry compressed air.

(3) If the base metal is exposed, touch-up as required with primer (PWC13-003).

7. Inspection/Check

A. Inspection of Propeller Thrust Bearing Cover Assembly

(1) Examine thrust bearing cover seal cavity for burrs, nicks and other surface damage. Light stoning of damaged surfaces is acceptable. Corrosion may be removed (Ref. Para. 8.).

B. Inspection of Oil Seal Retaining Ring or Ring Halves

(1) Examine oil seal retaining ring or ring halves for nicks, burrs and surface damage. Light stoning of damaged surfaces is acceptable.

(2) Check integrity of Post-SB1381/Post-SB1652 felt strip inserts in oil seal retaining ring halves . If worn or damaged, retaining ring halves must be shipped to an approved overhaul facility for felt strip insert replacement.

C. Inspection of Oil Seal Runner

(1) Examine visible area of seal runner for nicks, burrs and other surface damage caused by tools during removal of seal. No repair is acceptable. Ship power output section of engine to an approved overhaul facility for replacement of defective seal runner.

NOTE: If applicable, check the integrity of the sealant (PWC09-001) between the forward lip of the oil seal runner and propeller shaft. Replace if necessary.

(2) Examine the propeller shaft oil seal contact area on seal runner for wear (grooving). If concentrated wear is present, installation of an alternate offset propeller shaft oil seal should be considered. This offset seal will contact the runner at a different location (Ref. IPC).

D. Inspection of Scavenge Oil Strainer Assembly

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(1) Examine strainer element for blocked passages and for leakage between element (6, Fig. 203) and plug (4). Clean as necessary (Ref. Para. 6.A.).

(2) Remove plug (4) and preformed packing (5); check condition of preformed packing, then discard.

(3) Examine strainer element seating surfaces for nicks, burrs and other surface damage caused during removal and installation. Light stoning of damaged surfaces is acceptable.

(4) Replace strainer element assembly if cracks are detected.

(5) Examine threads of bolt and, if plug is removed, internal threads for deformation. Chase threads using suitable die or tap as appropriate. Remove all metal chips.

E. Inspection of Magnetic Chip Detector

NOTE: Whenever a reduction gearbox chip detector is inspected, the main oil filter should also be inspected. Any foreign material found on the chip detector or in the main oil filter should be identified before further inspection/operation. For detailed debris analysis refer to Chapter 70-00-00, MAINTENANCE PRACTICES.

CAUTION: WHEN AN ENGINE IS REMOVED DUE TO METAL CONTAMINATION OF THE OIL SYSTEM AND TO AVOID CONTAMINATION OF A NEWLY INSTALLED ENGINE, ALL AIRFRAME COMPONENTS ASSOCIATED WITH THE ENGINE OIL SYSTEM - INCLUDING PROPELLER, PROPELLER GOVERNORS, TUBES AND HOSES - SHOULD BE FLUSHED ACCORDING TO AIRCRAFT MAINTENANCE MANUAL. AIRFRAME OIL COOLERS SHOULD BE REPLACED WITH NEW COMPONENTS. ORIGINAL ENGINE EQUIPMENT SUCH AS OIL-TO-FUEL HEATERS AND GOVERNORS SHOULD BE RETURNED WITH THE ENGINE.

(1) Remove chip detector (Ref. Para.).

NOTE: Every time a chip detector is removed for inspection, the oil filter element should also be inspected (Ref. 79-20-02 Inspection/Check).

(2) If only a single chain of magnetic particles, consisting of slivers or fuzz type material or non-magnetic material, bridges the gap between poles of the chip detector, clean the chip detector and continue operation. Recheck both chip detector and oil filter within the next 15 flight hours.

NOTE: Clean poles on chip detector by wiping with a lint-free cloth; discard cloth after cleaning.

(3) When more than 10 small, nugget shaped, magnetic pieces are found, the engine should be returned to an approved overhaul facility for inspection.

(4) When a small amount of fuzz made up of powdered material is evident, clean detector, reinstall and recheck for electrical continuity after 10 hours.

(5) When powdered fuzz is evident a second time, but the quantity has not increased, clean detector, reinstall and recheck after a further 10 hours.

(6) When powdered fuzz is evident a third time in similar or greater quantities, the engine should be returned to an approved overhaul facility for inspection.

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(7) Detailed Inspection:

(a) Examine threads for damage. Chase threads as required. Replace detector if threads are badly damaged or crossed.

(b) Examine witness stripes on the detector housing and electrical connector. Movement of the connector relative to the housing may short the circuit's internal connections and give false indications. This condition is cause for rejection of the detector.

NOTE: Certain chip detectors incorporate a splined locking arrangement which prevents movement, therefore do not require witness stripe identification.

(c) Connect the ohmmeter across the detector output terminals and place a suitable steel bar across the magnet poles. Circuit continuity must exist.

(d) Check the output terminal pins and the condition of the potting compound. If the pins are loose or if the compound is cracked, reject the detector.

F. Inspection of Gearbox Rear Housing (Heavy Maintenance Only)

(1) If the exhaust duct assembly has been remove, examine the reduction gearbox rear case:

(a) Examine the exposed surface of the reduction gearbox rear case for carbon, oil wetting, etc.

(b) After cleaning (Ref. Para. 6. C.), examine the rear case surface for damage, corrosion, galling, pitting and the integrity of the protective coating.

(c) If the reduction gearbox rear case shows evidence of corrosion or pitting, return the power section to an approved overhaul facility for replacement of the reduction gearbox rear case.

G. Inspection of Propeller Shaft (Ref. Fig. 206)

(1) Inspect Area B of the propeller shaft for damage as follows:

(a) Isolated surface damage up to 0.003 in. deep and minimum 0.010 in. apart is acceptable.

NOTE: The anti-corrosion coating can be up to 0.004 in. thick. For propeller shaft with such coating, you can remove the coating locally in order to have a better view of the damage. Refer to the anti-corrosion coating repair per Para. 8.C.

(2) Inspect other surfaces as follows:

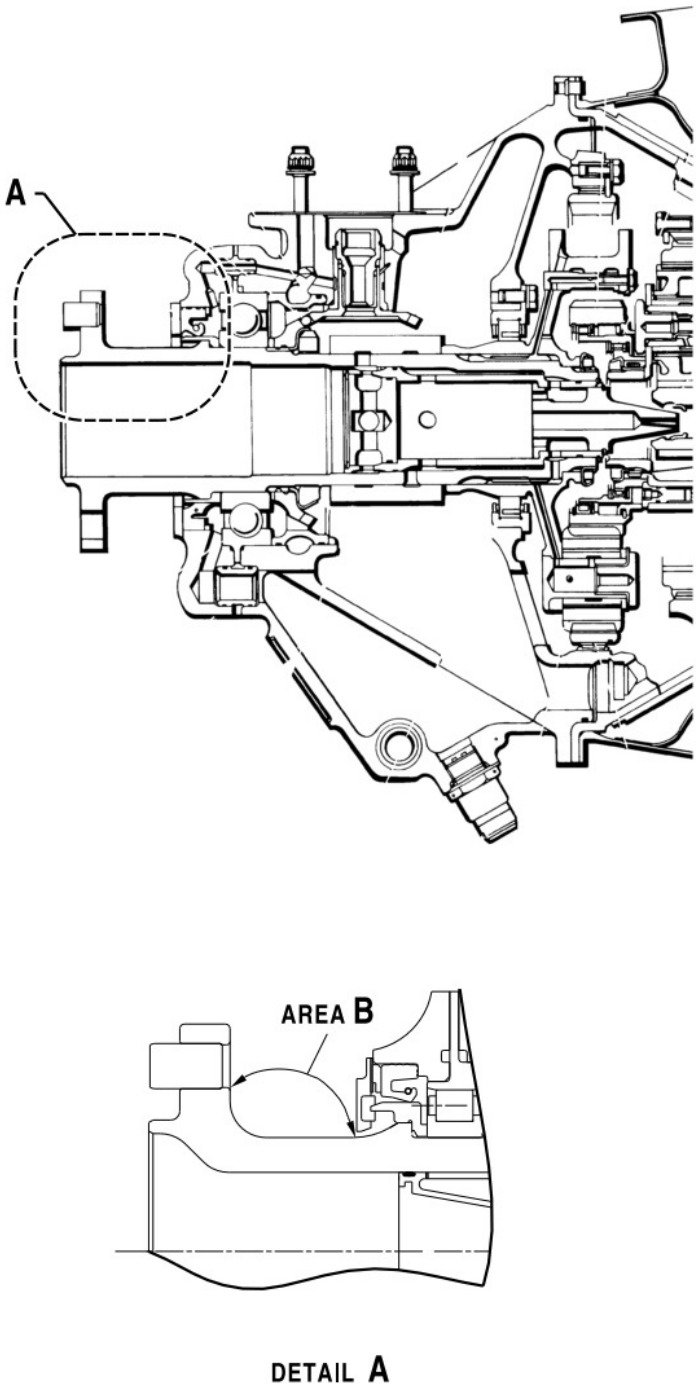
(a) Local surface damage on flange and pins up to 0.005 in. is acceptable.

(b) Local surface damage on remaining surfaces up to 0.002 in. deep is acceptable.

(3) If the propeller shaft has an anti-corrosion coating, perform local paint touch-up on exposed surfaces per Para. 8.C. as required.

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Figure 206 Inspection of Propeller Shaft



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8. Approved Repairs

A. Replacement of Accessory Drive Seals from Propeller Gearbox

(1) Remove accessory unit, if applicable from mounting pad on reduction gearbox.

(2) Remove seal and seal carrier (Ref. 72-60-00).

(3) Install replacement seal (Ref. 72-60-00).

(4) Reinstall accessory unit, as applicable.

B. Repair of Thrust Bearing Cover (Corroded Seal Cavity)

NOTE: Mask bearing and propeller shaft, to prevent ingress of debris.

(1) Remove corrosion from seal cavity contact surface using abrasive cloth (PWC05-061).

(2) Clean area with solvent (PWC11-027).

(3) Apply adhesive (PWC08-002) at corroded surface following manufacturer's instructions. Remove excess adhesive.

(4) Install seal element (Ref. Para. F.).

(5) Allow adhesive to cure following manufacturer's instructions.

C. Propeller Shaft Anti-corrosion Coating (Ref. Fig. 207)

(1) Mask the propeller shaft bore, the runner, seal area and all the surrounding surface.

(2) Install heat insulation blanket over the propeller thrust bearing cover area.

(3) Use abrasive cloth (PWC05-061) to remove any corrosion and/or stains on the propeller shaft. If required, use a abrasive paper 600-800 grit (PWC05-270) to locally remove the coating.

WARNING: REFER TO MANUFACTURER'S MATERIAL SAFETY DATA SHEETS FOR CONSUMABLE MATERIALS INFORMATION SUCH AS: HAZARDOUS INGREDIENTS, PHYSICAL/CHEMICAL CHARACTERISTICS, FIRE, EXPLOSION, REACTIVITY, HEALTH HAZARD DATA, PRECAUTIONS FOR SAFE HANDLING, USE AND CONTROL MEASURES.

(4) Clean the area surface to touch-up with isopropyl alcohol (PWC11-014B) commercial grade.

(5) Use a fine brush, apply a thin and even coat of paint 0.0015 to 0.002 in. thick as per one of the following options:

WARNING: REFER TO MANUFACTURER'S MATERIAL SAFETY DATA SHEETS FOR CONSUMABLE MATERIALS INFORMATION SUCH AS: HAZARDOUS INGREDIENTS, PHYSICAL/CHEMICAL CHARACTERISTICS, FIRE, EXPLOSION, REACTIVITY, HEALTH HAZARD DATA, PRECAUTIONS FOR SAFE HANDLING, USE AND CONTROL MEASURES.

(a) SermeTel 984 coating (PWC07-062A): Mix the aluminium coating base with the aluminium coating catalyst in the following proportion: 100 parts base to 17 parts

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catalyst by volume.
(or)

(b) Alseal 500 coating (PWC07-062B) or Alseal 519 coating (PWC07-062C): Mix the aluminium coating base with the aluminium coating catalyst in the following proportion: 100 parts base to 1.6 parts catalyst by volume.

(6) Air dry for 10 to 15 minutes. Cure the coating with heat lamp set at 190°C (375°F) minimum for 90 to 120 minutes.

(7) Use a soft cloth and manually burnish the coated surface.

(8) Use a strip of masking tape (PWC05-283) and check the coating for adhesion.

WARNING: REFER TO MANUFACTURER'S MATERIAL SAFETY DATA SHEETS FOR CONSUMABLE MATERIALS INFORMATION SUCH AS: HAZARDOUS INGREDIENTS, PHYSICAL/CHEMICAL CHARACTERISTICS, FIRE, EXPLOSION, REACTIVITY, HEALTH HAZARD DATA, PRECAUTIONS FOR SAFE HANDLING, USE AND CONTROL MEASURES.

(9) Use a fine brush to apply a thin, even coat of epoxy enamel paint (PWC07-015). Use color required to match existing coating.

(a) The mix ratio is one part of catalyst to one part of base.

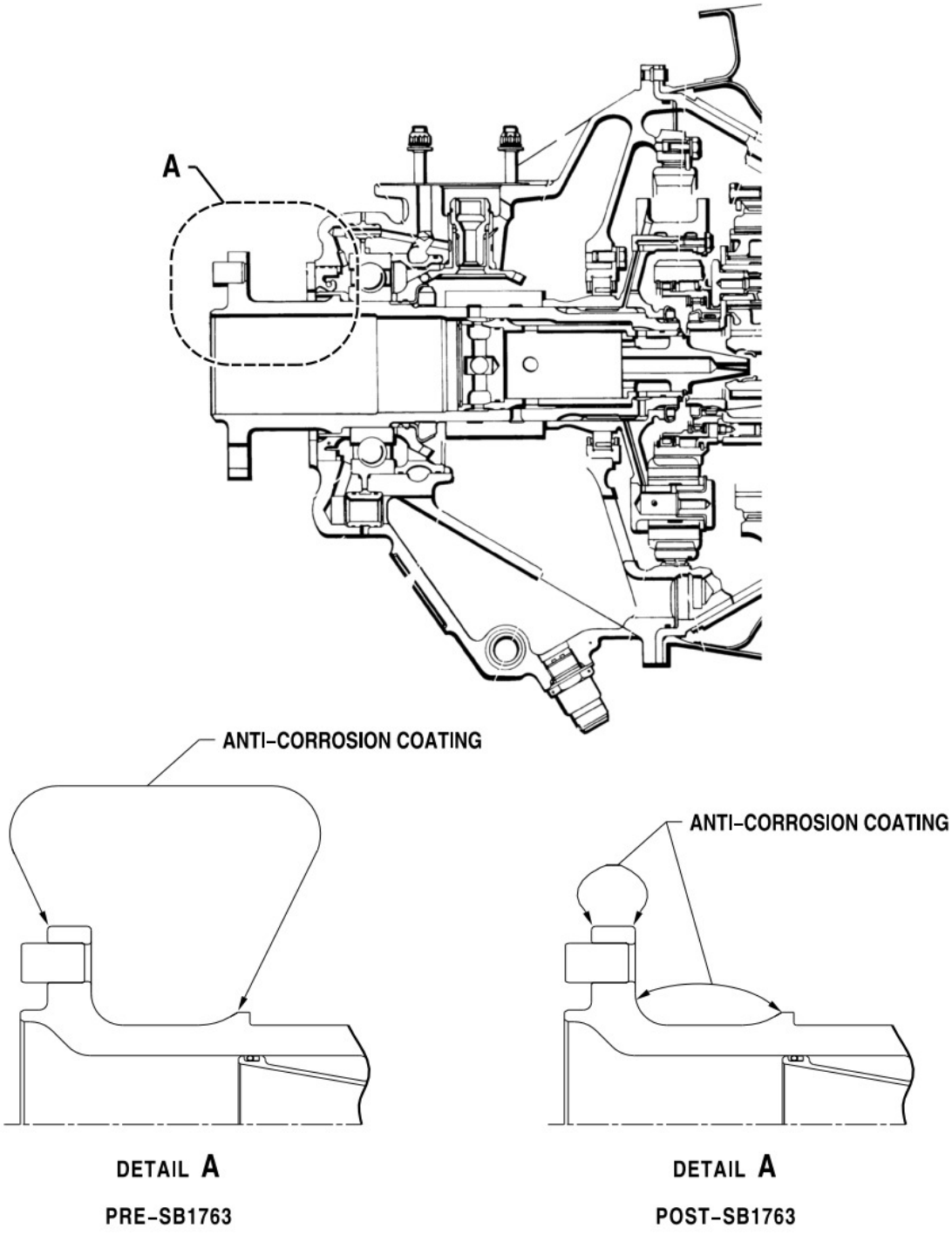
(10) Air dry for 15 to 20 minutes. Cure the top coat with heat lamp set at 80°C (175°F) minimum for 4 hours.

(11) Use a strip of masking tape (PWC05-283) and check the coating for adhesion.

(12) Remove all masking and heat insulation blankets.

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Figure 207 Propeller Shaft Anti-corrosion Coating - Local Restoration



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D. Repair of Corrosion - Front and Rear Reduction Gearbox Housing

NOTE: Corrosion must not be deeper than 0.010 in. and must not cover an area greater than 30 percent of the total surface of the front and rear RGB housing.

(1) Apply a suitable covering material around the area to be repaired.

(2) Clean the surface of the area to be repaired with a swab soaked in isopropyl alcohol (PWC11-014) and crocus cloth (PWC05-061).

(3) Remove all traces of corrosion (magnesium oxide) using a suitable steel brush, crocus cloth (PWC05-061), abrasive cloth (PWC05-101), file or grit paper. Remove all traces of debris using a vacuum cleaner.

(4) Flush the area with clean water at room temperature.

(5) Dry the area with clean compressed air at 29 psig.

WARNING: REFER TO THE MATERIAL SAFETY DATA SHEETS BEFORE YOU USE THESE MATERIALS FOR INFORMATION SUCH AS HAZARDOUS INGREDIENTS, PHYSICAL/CHEMICAL CHARACTERISTICS, FIRE, EXPLOSION, REACTIVITY, HEALTH HAZARD DATA, PRECAUTIONS FOR SAFE HANDLING, USE AND CONTROL MEASURES. SOME OF THESE MATERIALS CAN BE DANGEROUS. YOU CAN GET THE DATA SHEETS FROM THE MANUFACTURERS OR THE SUPPLIERS OF THESE MATERIALS.

(6) Prepare a chrome pickle solution (Ref. Chap. 70-00-00, MAINTENANCE PRACTICES, Touch-Up Solution).

(7) Using a swab or brush, apply the chrome pickle solution, at a temperature of 17° to 29°C (55° to 85°F), to the prepared surface for 30 to 45 seconds.

NOTE: Repeat the application frequently to make sure that the affected surface is continually wet with the solution.

(8) Swab the area with clean water until successive swabs are no longer stained yellow.

(9) Dry the area with local heating (air dry or a heat gun at a low setting).

(10) Clean the affected area with a rag soaked in clean water.

(11) Dry the area with clean, compressed air at 29 psig.

(12) Apply two coats of primer (PWC13-001).

NOTE: The primer can be diluted with 10% solvent.

(13) Allow the primer to air dry for eight hours before applying enamel paint. Use compressed air at 29 psig to accelerate the drying time.

(14) Apply three to four coats of enamel (PWC05-037) to the primed surface. Allow the surface of the enamel to become tacky (approximately 15 minutes) between each coat. The final coat of enamel must dry for 24 hours before returning the engine to service.

NOTE: Drying time for the primer and paint can be reduced with the use of a heat gun at a low setting. Refer to the manufacturers instructions.

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